

KARTHIKEYA SHARMA

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Education

Georgia Institute of Technology

08/2023 – 05/2025

Master of Science in Electrical and Computer Engineering

GPA: 3.77/4

SRM Institute of Science and Technology

07/2019 – 05/2023

Bachelor of Technology in Electronics and Communication Engineering

GPA: 9.83/10

Technical Skills

Coding: C/C++, CUDA C, *Scripting* (Python, Bash, TCL), *RTL* (Verilog, SystemVerilog), High-Level Synthesis

Tools: Vivado, Vitis HLS, Modelsim, Verdi, VCS

Agents: Claude Code, Open Code

Protocols: AXI (AXI-MM, AXIS, AXI-Lite), QDMA, PCIe

Libraries: OpenMP, OpenMPI, NeMO Guardrails

Coursework: Advanced Computer Architecture, Hardware-Software Co-Design for ML Systems, Parallel Programming for FPGA, Advanced Programming Techniques, Machine Learning, VLSI Design, Microprocessors & Microcontrollers.

Work Experience

Hardware Engineer

07/2025 – Present

MangoBoost, Inc.

Bellevue, WA

System Architecture & Implementation Team | Hardware Development

07/2025 – 01/2026

- Served as core architect of Data Plane Studio (DPS) for the **Mango BoostX 400G DPU**, enabling scalable reuse of a single hardware architecture across multiple use cases: ROCEv2, NVMe-over-RDMA, & NVMe-over-TCP Initiator.
- Contributed to PCIe and DMA-centric feature development, resolved critical functional, timing, and performance bottlenecks—including root-causing DMA transaction stalls to NUMA topology and stabilizing reset-tree architecture—to ensure the delivery of DPS for alpha-verified milestones.
- Developed an encrypted, IP-protected package for DPS to secure and obfuscate proprietary hardware designs, accelerating early-adopter onboarding by delivering high-performance demo examples and comprehensive user guides.
- Built scalable testbenches for demo examples using cocotb, featuring Python-based BFM and custom test cases to empower users to independently validate design functionality in simulation.
- Built automation scripts to streamline DPU testing, enhanced CI/CD workflow for sanity testing—covering synthesis/implementation checks, functional verification, and network performance test.

Solution Architecture Group | Workload Analysis & Performance Optimization

02/2026 – Present

- Investigated SOTA rack-scale AI infrastructure, software stacks (NVAIE, AMD AI), & microservices (e.g., guardrails) to evaluate existing deployment methodologies, assessing how on-the-fly DPU processing can optimize system workloads.
- Profiled execution traces of coding agent sessions (Claude Code, OpenCode) to characterize agentic workflows, identifying high-frequency tool-call volumes as a primary overhead for CPU.
- Developed a LangChain-based agentic flow integrated with NeMo Guardrails, serving models w/ vLLM (1 primary LLM, 2 guardrail models) across 3 servers to benchmark latency overhead and throughput degradation.
- Engineered a sandboxed testing environment and AI benchmarker to securely execute coding agents and evaluate the performance of inference frameworks (e.g., vLLM, llama.cpp).
- Evaluated additional use cases for expanding DPUs in scale-out system architectures: NVMe-over-RDMA for storage disaggregation (KV Cache offloading), KV Cache compression, and accelerating guardrails on DPUs.

Research Experience

Cryptonite: Scalable Accelerator Design for Cryptographic Primitives | [Paper](#)

01/2024 – 05/2025

- Built Cryptonite, a HLS-based design-space-exploration toolchain that compiles formally verified straight-line C from Fiat-Cryptography into **hundreds of correct-by-design Pareto-optimal** designs for elliptic-curve primitives.
- Designed an **analytical QoR estimator** that predicts latency/DSP usage during pre-synthesis making exhaustive pragma exploration tractable.
- Achieved **89% lower resource usage** and **54% lower latency** compared to the straight-line code baseline.

APEX: Automated Platform for EXploring Collective Communication Systems | [Code](#)

04/2024 – 12/2024

- Prototyped **APEX**, an automated tool-chain for analyzing network configurations and collective algorithms for ML algorithms via Design Space Exploration (DSE), revealing 64% of the workload bandwidth and topology dependent.
- Streamlined collective algorithm synthesis by translating TACOS outputs to MSCCLang IR and Chakra Execution Traces, bypassing ASTRA-Sim's system layer reducing engineering effort by 10×.
- Enabled heterogeneous topology translation and an interactive visualizer for analysis of GPU network collective flows and patterns for congestion and switch modeling.

Teaching Experience

Graduate Teaching Assistant (CS 6340 : Software Analysis and Testing)	08/2024 – 12/2024
Dept. of Computer Science, Georgia Institute of Technology	Atlanta, GA
- Delivered a class session for 35 students on SVF framework, including a live tutorial & debugging strategies. - Conducted weekly office hours mentoring students on assignments and final project. Assisted in developing 21 test cases for the project and evaluated students' implementation strategies for technical rigor.	

Publications

- Xu, H., Chen, W., Dixon, J. Z., Maharaj, D. S., Sharma, K. M., Audier, X., Cicerone, M. T. Ultra-high Information-content Chemical Imaging with Broadband Coherent Anti-Stokes Raman and Two-photon Fluorescence Lifetime Microscopy. J. Vis. Exp. (224), e68845, doi:10.3791/68845 (2025). (Published Oct 7, 2025) K. S. Maheswaran, C. Bossut, A. Wanna, Q. Zhang and C. Hao, "Cryptonite: Scalable Accelerator Design for Cryptographic Primitives and Algorithms," 2025 IEEE 36th International Conference on Application-specific Systems, Architectures and Processors (ASAP), Vancouver, BC, Canada, 2025, pp. 17-24, doi: 10.1109/ASAP65064.2025.00013. (Published July 31, 2025) K. S. M., A. Rao, K. Kumar and T. R. Rao, "Terahertz Imaging for Aerospace Applications," 2023 International Conference on Wireless Communications Signal Processing and Networking (WiSPNET), Chennai, India, 2023, pp. 01-05, doi: 10.1109/WiSPNET57748.2023.10134245. (Published May 30, 2023)	
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Conference Presentations

Represented MangoBoost at FCCM 2026, presenting the BoostX DPU and showcasing its hardware capabilities through a live technical demonstration during Demo Night.

Projects

Federated training and profiling of large-scale ML model <i>Python</i>	03/2024 - 04/2024
- Implemented single and distributed GPU training (2 GPUs) of a GPT-like (gpt2-large architecture) model on NVIDIA RTX 6000 server using Megatron-LM, utilizing tensor, data, and pipeline parallelism. - Profiled training with TensorBoard showing tensor parallelism achieving 1.84× speedup over pipeline parallelism.	
Performance-Aware Accelerator Design for ML workloads <i>Python</i>	02/2024 - 03/2024
- Conducted DSE using Scale-Sim to optimize systolic array configurations for LeNet CNN, achieving 57% average mapping efficiency for operators while evaluating performance trade-offs across dataflow configurations. - Developed a mapping strategy (mapping efficiency 84.375%) to mitigate faulty processing elements in a weight-stationary 3x3 systolic array, optimizing GEMM computation latency (54 cycles).	
MIPS CPU and CMP memory simulator <i>C++</i>	09/2023 - 12/2023
- Developed a 5-stage (8-pipe) MIPS CPU simulator with Out-of-Order execution, GShare Branch Prediction, and optimized cache hierarchy, reducing CPI by 50% and latency by 10%. - Engineered a dual-core simulator with L1, L2-shared caches (LRU & random eviction) and DRAM, optimized L2 utilization via SWP & DWP for a 0.85% performance gain.	